

21.0 PAINTING OPERATIONS

Policy

Painting and paint removal present hazards requiring effective controls. Hazards include exposure to toxic materials and flammable or explosive mists, particulates, and vapors.

Inhalation of mists and vapors from nearly all paints, solvents, thinners, cleaning chemicals, strippers, and epoxies can be injurious depending upon the agent's toxic characteristics and the amount and method of exposure. Further, many can physically injure the skin and eyes, or be absorbed through the skin.

Potential physical and health hazards can be effectively controlled by appropriate work procedures, controls, facility design, protective clothing, and equipment.

Procedures

Pressure Equipment

Pressure equipment used in painting operations is hazardous because of the compressed air component; therefore, the Supervisor shall assure that spray painting equipment is in serviceable condition.

On all air-type spraying equipment a pressure regulator valve shall be installed in the air line between the compressor and painting equipment. A pressure relief valve and a pressure gauge shall be installed between the pressure regulator and pressurized paint containers and/or spray guns. Pressure relief valves shall be set to open at pressures not more than 10 pounds above the required working pressure.

Other Equipment

Painter's ladders, scaffolds, and other equipment shall be inspected prior to use to be certain they are in safe condition.

Paint Mixing

Paint mixing shall be done in designated, adequately ventilated rooms constructed of fire-resistant materials. All sources of ignition shall be prohibited in mixing areas. All electrical fixtures or equipment in or within 20 feet of designated paint preparation areas shall meet the requirements of the National Electrical Code (NFPA #70) for Class I Division 2 locations.

Housekeeping

Good housekeeping is essential to safe operations in paint shops. Paint rooms, booths, etc., shall be kept clean with equipment stored in a proper and orderly manner. All solvent or paint soiled rags shall be placed in approved self-closing metal containers plainly marked to indicate the contents. At the end of each day, these containers shall be emptied or removed to an approved location for pickup and disposal.

Health

Personnel Exposures-

There is a wide application of organic solvents in painting. All organic solvents have some effect on the central nervous system and the skin. The principal modes of personnel exposure are inhalation of vapors and skin contact and absorption. Personnel engaged in painting operations should review Material Safety Data Sheets (SDS) in order to acquaint themselves with the properties and hazards of the solvents that are used. Skin contact with solvents may cause dermatitis, ranging in severity from a simple irritation to actual damage to the skin.



SAFETY AND HEALTH PROGRAM

Protective Equipment-

Personnel engaged in painting and paint removal shall wear protective clothing, respiratory devices if required, and appropriate face, eye, and hand protection. Eye or face protection is required during scraping or paint preparation (abrasive techniques). Clothing shall be changed, as needed, to minimize body contamination.

Respiratory Protection-

The Safety and Health Manager should be consulted for specific advice on respiratory protection required for specific painting activities.

.Personal Hygiene-

The hands and face shall be kept clean, clothes shall be changed when contaminated and hands and soiled objects shall be kept out of the mouth.

No food or drink shall be brought into, or consumed, in paint shops. Personnel shall wash their hands prior to smoking or consuming food. Air and Water Pollution

Pollution Prevention-

Painting and paint removal operations can cause air and water pollution problems which can impact the local community. Liquid, solid, and gaseous waste products from painting and paint removal operations shall be disposed of in accordance with federal and state air, water, and solid waste pollution control laws and as specified by the Safety and Health Manager.

Spills-

All spills of flammable or combustible liquids shall be cleaned up promptly. With major spills, remove ignition sources, evacuate, and ventilate the area, and provide appropriate protective equipment to cleanup personnel. These liquids shall not be allowed to enter a confined space, such as a sewer, because of the possibility of an explosion.